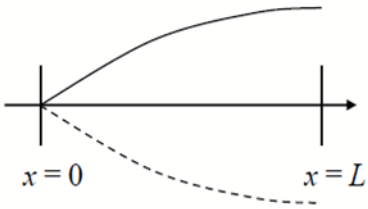
# Stehende Welle mit festem und losem Ende

a) Randbedingungen:

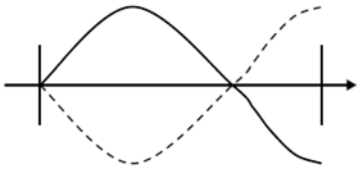
RB I) $s_0(x=0) = 0$ (Knoten)

RB II) $s_0(x=L) = \max.$ (Bauch)

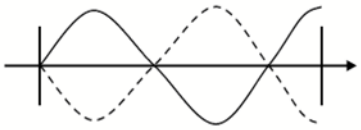
b)




Grundmode



1. Oberschwingung



2. Oberschwingung

c) allg. Ansatz $\boxed{s_0(x) = A \cdot \sin(kx) + B \cdot \cos(kx)}$

wg. RB I $\Rightarrow$ $s_0(x=0) = 0$

$\Rightarrow$ $\boxed{B = 0}$, da $\cos(0) = 1$ & $\sin(0) = 0$

wg. RB II $\Rightarrow$ $\sin(k \cdot L) = \pm 1$

$k \cdot L = \underbrace{\frac{\pi}{2} + n \cdot \pi}_{\text{hier wird der Sinus } \pm 1}$ f. $n \in \mathbb{N}$

$\Rightarrow$ $\boxed{k_n = \frac{\pi}{L}\left(n + \frac{1}{2}\right) \text{ mit } n \in \mathbb{N}}$

Grundmode $k_0 = \frac{\pi}{L} \cdot \frac{1}{2} = \frac{\pi}{0{,}2\,\text{m}} \cdot \frac{1}{2} \approx 7{,}85\,\text{m}^{-1}$

1. Oberschwingung $k_1 = \frac{\pi}{L}\left(1 + \frac{1}{2}\right) = \frac{\pi}{0{,}2\,\text{m}} \frac{3}{2} \approx 23{,}6\,\text{m}^{-1}$

Wellenlänge $\lambda_n$: $\lambda_n = \frac{2\pi}{k_n}$

$$\Rightarrow \boxed{\lambda_n = \frac{2L}{n+1/2} \quad \text{mit} \quad n \in \mathbb{N}}$$

Grundmode: $\lambda_0 = \frac{2L}{1/2} = 4L = 0{,}8\ m$

1. Oberschwingung $\lambda_1 = \frac{2L}{3/2} = \frac{4}{3}L = 0{,}27\ m$

Frequenz $f_n$: $c_L = \lambda_n \cdot f_n \Rightarrow f_n = \frac{c_L}{\lambda_n}$

$$\Rightarrow \boxed{f_n = \frac{c_L}{2L}(n+1/2) \quad \text{mit} \quad n \in \mathbb{N}}$$

Grundmode: $f_0 = \frac{c_L}{2L} \cdot 1/2 = \frac{c_L}{4L} \approx \frac{340\ m/s}{4 \cdot 0{,}2\ m} \approx 425\ Hz$

1. Oberschwingung $f_1 = \frac{c_L}{2L} \cdot 3/2 = \frac{3}{4}\frac{c_L}{L} \approx \frac{3 \cdot 340\ m/s}{4 \cdot 0{,}2\ m} \approx 1275\ Hz$